CTS Furthest-From-Opportunity Model: Technical Details

This document describes the OLS regression model used to predict the share of selected CTS scholars who are 'furthest from opportunity' (both first-generation college students AND from families earning at or below 65% of the HUD Median Family Income).

Model Specification

Furthest Share (Selected) = B0 + B1 * ln(N Furthest Applicants / N Slots) + error

The natural log transform is used because the relationship exhibits diminishing returns: when there are already many furthest applicants per slot, adding more has a smaller marginal effect on the selected share.

Fitted Parameters

Intercept (B0): 0.655583
Slope (B1): 0.231212
R-squared: 0.8948
p-value (slope): 0.001268
Number of observations: 7
Residual Std Error: 0.026811

Inverse Formula (Target Setting)

To find the number of furthest-from-opportunity applicants needed:

N Furthest Needed = N Slots x exp((Target Share - B0) / B1)

N Furthest Needed = N Slots x exp((Target Share - 0.6556) / 0.2312)

Good / Better / Best Estimates

Good: The number of furthest-from-opportunity applicants needed so that the best-fit regression line predicts the target share. This is the minimum needed if the historical relationship holds exactly.

Best: The number of furthest-from-opportunity applicants needed so that the lower bound of the 95% prediction interval equals the target share. This provides a buffer against year-to-year variability. Even in an unfavorable year, you would still expect to meet the target.

Better: The midpoint between Good and Best.

Prediction Intervals

The 95% prediction interval accounts for both estimation uncertainty (how well we know the regression line) and residual variance (natural year-to-year scatter). The interval width varies with the predictor value, being narrowest near the center of the observed data and wider at the extremes.

Historical Data

Year	Applicants	Selected	Furthest	Share	Ratio
2019-20	1,434	552	704	71.9%	1.28
2020-21	1,390	557	643	67.5%	1.15
2021-22	1,409	634	626	65.3%	0.99
2022-23	1,565	1,205	750	52.9%	0.62
2023-24	1,725	1,301	829	52.2%	0.64
2024-25	1,770	1,275	848	56.8%	0.67
2025-26	1,728	1,275	852	60.9%	0.67

Caveats

- The model is fit on 7 observations (one per application year). Standard errors should be interpreted cautiously.
- The model assumes the program's selection preference for furthest scholars remains approximately stable over time.
- Extreme target values (near 0% or 100%) produce unreliable predictions.
- The applicant pool composition is exogenous. The model indicates how many furthest applicants are needed but does not control whether they apply.